KTH Royal Institute of Technology

SF2527 Numerical Methods for Differential Equations I

Computer Exercise 1

Ordinary Differential Equations and Elliptic Equations

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1 Runge–Kutta accuracy and stability

The Runge–Kutta method used in Parts 1 and 2 is

$$k_1 = f(t_n, u_n), \qquad k_2 = f(t_n + h, u_n + hk_1), \tag{1.1}$$

$$k_3 = f\left(t_n + \frac{h}{2}, u_n + \frac{h}{4}k_1 + \frac{h}{4}k_2\right), \qquad u_{n+1} = u_n + \frac{h}{6}(k_1 + k_2 + 4k_3). \tag{1.2}$$

It is explicit, has three stages and uses one step at a time.

1.1 Landau–Lifshitz system

Let

$$a = \frac{1}{4}\begin{bmatrix} 1 \\ \sqrt{11} \\ 2 \end{bmatrix}, \qquad C = \frac{1}{4}\begin{bmatrix} 0 & -2 & \sqrt{11} \\ 2 & 0 & -1 \\ -\sqrt{11} & 1 & 0 \end{bmatrix}.$$

Since $Cm = a \times m$, the ODE is

$$m' = Am, \qquad A = C + \alpha C^2, \qquad \alpha = 0.07, \qquad m(0) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^{\mathsf{T}}.$$

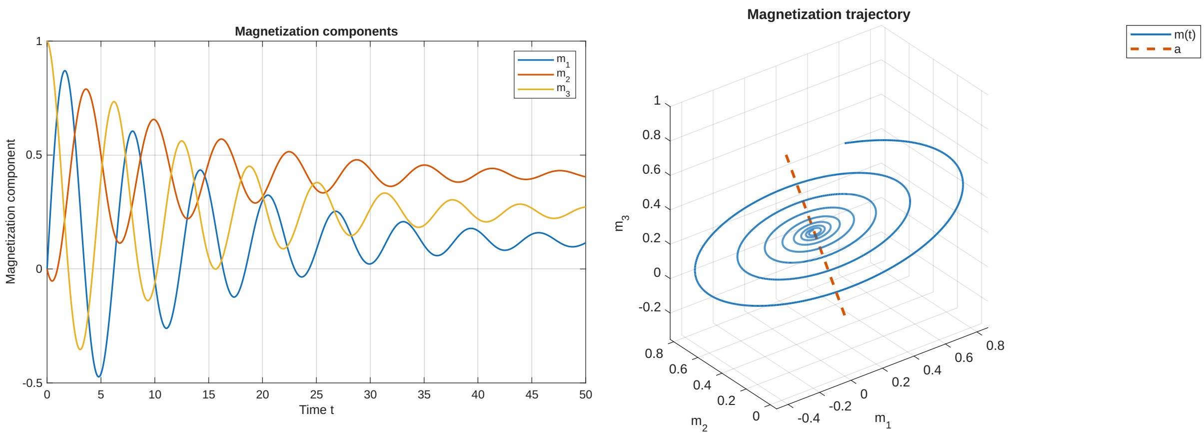

Figure 1: Magnetization components and trajectory for $0 \le t \le 50$ with $h = 0.01$.

The value $a^{\mathsf{T}}m(t) = a^{\mathsf{T}}m(0) = 1/2$ stays constant. Therefore $m(t)$ stays in a plane perpendicular to a. The part perpendicular to a goes to zero, so

$$m(t) \longrightarrow (a^{\mathsf{T}}m(0))a = \frac{1}{2}a,$$

This is also what we see in the plots.

1.2 Order of accuracy

For $N = 50, 100, 200, 400, 800, 1600$, let $h = 50/N$ and

$$d_N = \|\widetilde{m}_N(50) - \widetilde{m}_{2N}(50)\|_2 .$$

The slope in the log–log plot is

$$\boxed{p = 2.994893},$$

which is close to 3. The method is therefore third order.

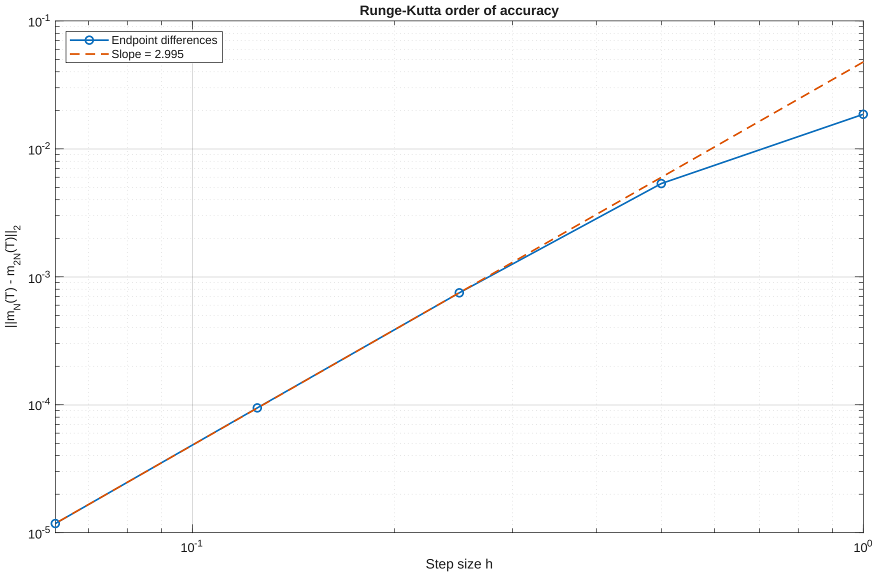

Figure 2: Endpoint differences and fitted order.

1.3 Stability limit

The eigenvalues of A are

$$\lambda_1 = 0, \qquad \lambda_{2,3} = -0.07 \pm i.$$

The stability function is

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}.$$

We need $|R(h\lambda_j)| \leq 1$ for every eigenvalue. The zero eigenvalue gives $R(0) = 1$. For $\lambda = -0.07 + i$, we solve

$$|R(h\lambda)| - 1 = 0$$

with `fzero` on $[2, 2.5]$. This gives

$$\boxed{h_0 = 2.055596}.$$

The test with $h = 2$ is stable. The test with $h = 50/24 = 2.083333$ is unstable.

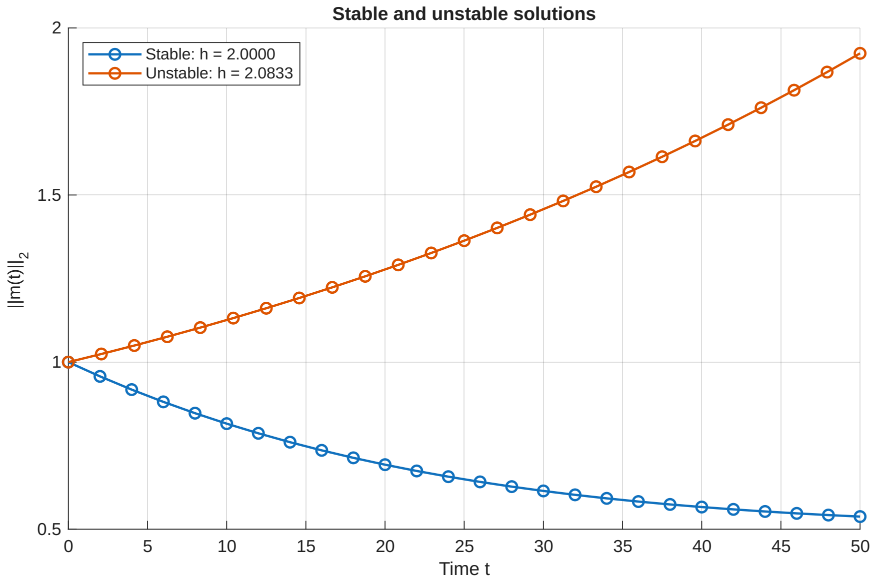

Figure 3: Stable and unstable solutions on opposite sides of h_0.

2 Robertson's stiff system

The concentrations $x = [x_A, x_B, x_C]^\mathsf{T}$ satisfy

$$x_A' = -r_1 x_A + r_2 x_B x_C, \tag{2.1}$$
$$x_B' = r_1 x_A - r_2 x_B x_C - r_3 x_B^2, \tag{2.2}$$
$$x_C' = r_3 x_B^2, \tag{2.3}$$

where

$$r_1 = 5.0 \times 10^{-2}, \qquad r_2 = 1.2 \times 10^{4}, \qquad r_3 = 4.0 \times 10^{7}, \qquad x(0) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^\mathsf{T}.$$

The sum $x_A + x_B + x_C$ should stay equal to 1, and it did so in our results.

2(a) Explicit Runge–Kutta on $[0, 10]$

On $0 \le t \le 10$, $h = 7.0 \times 10^{-4}$ gave a stable solution. In the normal plot, x_B is too small to see clearly. The log plot shows its fast change near $t = 0$.

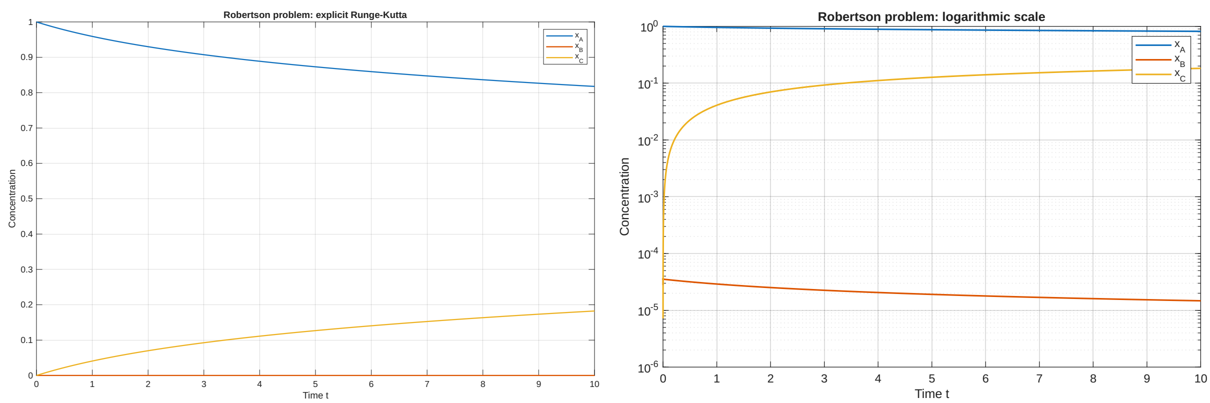

Figure 4: Explicit RK solution on $0 \le t \le 10$ with $h \approx 7.0 \times 10^{-4}$.

2(b) Jacobian eigenvalues and stability

The Jacobian is

$$J(x) = \begin{bmatrix} -r_1 & r_2 x_C & r_2 x_B \\ r_1 & -r_2 x_C - 2 r_3 x_B & -r_2 x_B \\ 0 & 2 r_3 x_B & 0 \end{bmatrix}.$$

For a small change e, the linearized equation is $e' \approx J(x(t))e$. Therefore we check $|R(h\lambda_j(t))| \leq 1$. One eigenvalue is zero. Along the computed solution,

$$\max_t |\lambda_j(t)| = 3365.924700.$$

The end of the RK stability region on the negative real axis gives

$$h_{\max} \approx 7.4652 \times 10^{-4},$$

which is close to the value found by testing. The large eigenvalue grows with time, so the allowed step size gets smaller.

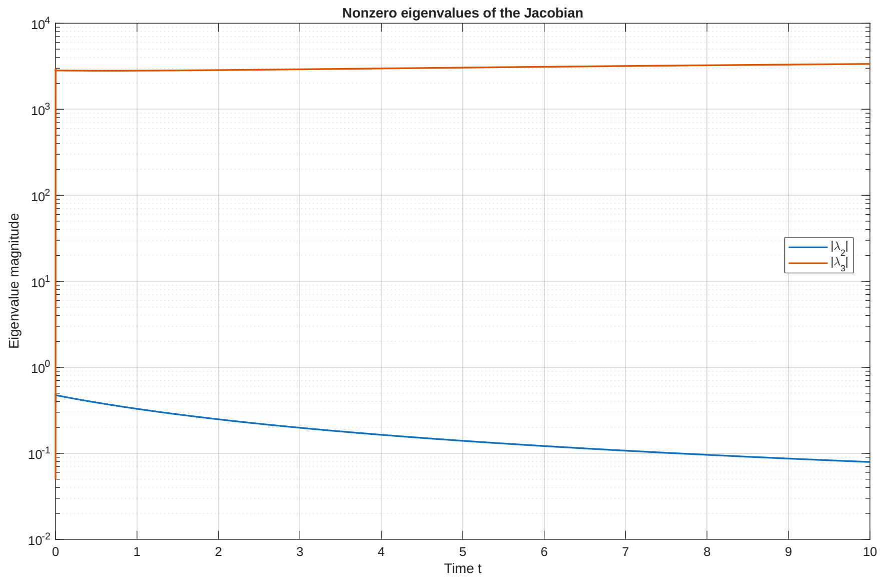

Figure 5: Magnitudes of the two nonzero Jacobian eigenvalues.

2(c) Explicit Runge–Kutta on $[0, 1000]$

For the long run, we stored only the current value of x.

Table 1: Explicit RK result at $t = 1000$.

Quantity	Value
h	2.8×10^{-4}
$x_A(1000)$	0.293414227164
$x_B(1000)$	$1.716342048384 \times 10^{-6}$
$x_C(1000)$	0.706584056494
Time	2.93 s

2(d) Implicit Euler

Implicit Euler satisfies

$$u_{n+1} = u_n + hf(u_{n+1}).$$

At each step, the supplied function solves

$$F(v) = v - hf(v) - u_n = 0$$

with Newton's method. Here v is the new solution value:

$$(I - hJ_f(v))d = -F(v), \qquad v \leftarrow v + d.$$

Runs with $h = 10$, 1 and 0.1 did not blow up. Large steps are stable, but not accurate. With $h = 0.1$, the IE and RK curves are close on $[0, 10]$. We used the same step size on $[0, 1000]$.

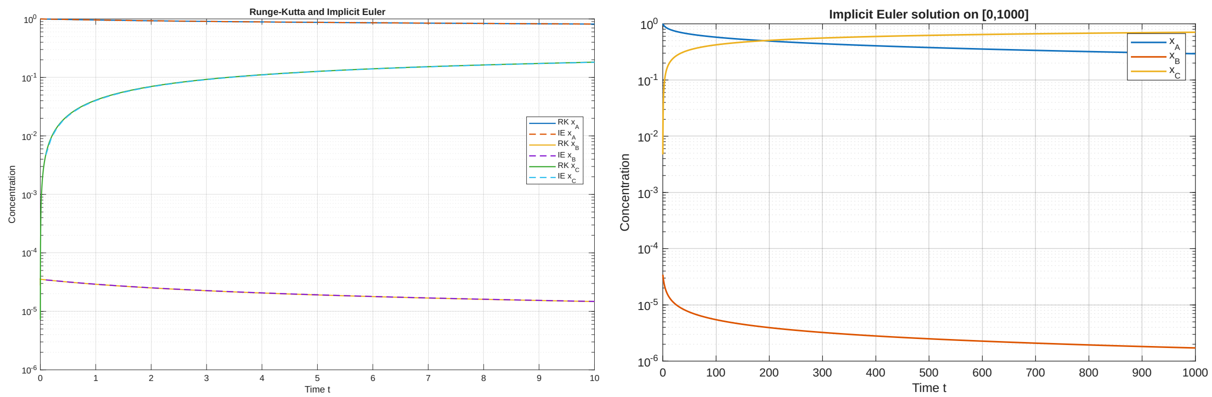

Figure 6: Short-time RK/IE comparison and the IE solution on $[0, 1000]$.

Note: Both the RK and implicit Euler solutions are plotted over $0 \le t \le 10$, but their curves overlap almost completely and are therefore difficult to distinguish.

2(e) Accuracy and efficiency

We used the given reference value at $t = 1000$ to calculate the errors. For the timing test, the IE code stores only the current value.

Table 2: Endpoint error and single-run computational time.

Method	h	Error	Time (s)
RK	2.8×10^{-4}	3.06×10^{-13}	2.93
IE	1	4.93×10^{-4}	0.014
IE	0.1	5.03×10^{-5}	0.048
IE	0.01	5.04×10^{-6}	0.409
IE	0.001	5.04×10^{-7}	4.04

The timings are influenced by both the number of steps and the work per step. For a fixed final time T, the number of steps is

$$N = \frac{T}{h}.$$

The RK method uses three right-hand-side evaluations per step, whereas each IE step requires several Newton iterations and a linear solve. An IE step is therefore more expensive, but IE can use a much larger step size on this stiff problem.

The methods also have different accuracy orders. In the asymptotic regime, their endpoint errors behave approximately as

$$e_{\mathrm{RK}}(h) \approx C_{\mathrm{RK}} h^3, \qquad e_{\mathrm{IE}}(h) \approx C_{\mathrm{IE}} h.$$

The IE values in the table display the expected first-order behavior: reducing h by a factor of ten reduces the error by approximately the same factor. For RK, stability forces the small step size $h = 2.8 \times 10^{-4}$, corresponding to about 3.57×10^6 steps on $[0, 1000]$. Its computed endpoint agrees with the given reference values to their stated precision. The reported 3.06×10^{-13} difference is therefore near the precision limit of the reference and should not be interpreted as an equally precise measurement of the true error.

IE is faster when an error around 10^{-4} or 10^{-5} is enough, since it can take large steps. The table places the crossover near 10^{-6}: IE takes 0.409 seconds at an error of 5.04×10^{-6}, while it takes 4.04 seconds at an error of 5.04×10^{-7}, compared with 2.93 seconds for RK. For smaller errors, IE needs many steps because it is only first order, and each step also needs a Newton solve. RK is therefore faster below an error of about 10^{-6}. Part 1 is not stiff, so IE has no useful stability benefit there. The higher-order RK method is a better choice for that problem.

3 One-dimensional convection–diffusion equation

We use the grid $z_j = jh$, where $h = 1/N$. Because T_0 is prescribed, the unknown vector is $U = [T_1, \dots, T_N]^\mathsf{T}$, and we assemble the sparse linear system $AU = q$. Centered second-order differences give, for $j = 1, \dots, N-1$,

$$\ell T_{j-1} + dT_j + rT_{j+1} = Q(z_j),$$

where

$$\ell = -\frac{1}{h^2} - \frac{v}{2h}, \qquad d = \frac{2}{h^2}, \qquad r = -\frac{1}{h^2} + \frac{v}{2h}.$$

Thus the first $N-1$ rows of A are tridiagonal. In the first row, the known boundary value is moved to the right-hand side, so

$$q_1 = Q(z_1) - \ell T_0.$$

At $z = 1$, we approximate the derivative by

$$T'(1) \approx \frac{3T_N - 4T_{N-1} + T_{N-2}}{2h}.$$

Substitution into the Robin boundary condition gives the final matrix row

$$T_{N-2} - 4T_{N-1} + (3 + 2h\alpha)T_N = 2h\alpha T_{\mathrm{out}}.$$

The code then solves the system once with `U = A\rhs` and returns $T = [T_0; U]$.

3(a) Grid refinement

Table 3: Temperature at the exact grid point $z = 0.5$.

N	h	$T_h(0.5)$
80	0.0125	258.430491144194
160	0.00625	258.621578905525
320	0.003125	258.669336283685

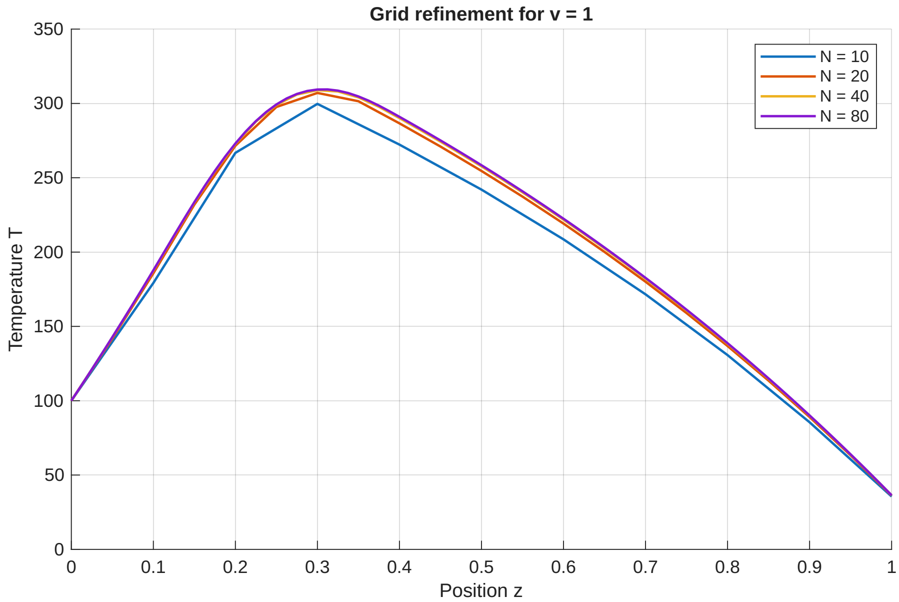

Figure 7: Solutions for $v = 1$ and $N = 10, 20, 40, 80$.

The three values give

$$p = \log_2\left(\frac{|T_{80}(0.5) - T_{160}(0.5)|}{|T_{160}(0.5) - T_{320}(0.5)|}\right) = \boxed{2.000440}.$$

This is very close to order 2. We evaluate Q at the grid points z_j and read $T(0.5)$ at $j = N/2$ on every grid.

3(b) Different velocities

Table 4: Grids used for the velocity comparison.

v	N	h
1	80	0.0125
5	100	0.0100
15	300	0.003333
100	2000	0.0005

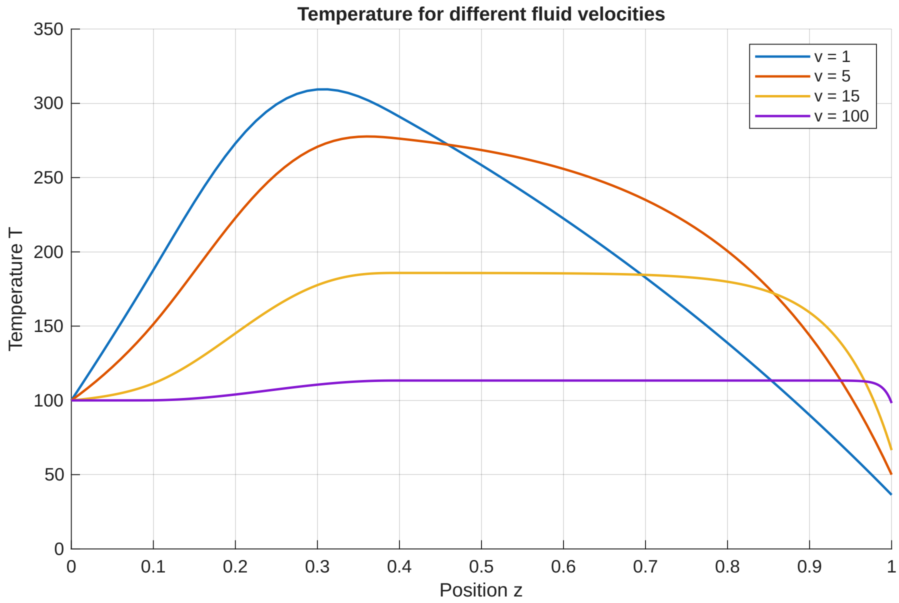

Figure 8: Temperature profiles for $v = 1, 5, 15, 100$.

When v is larger, the fluid moves the heat farther down the pipe. The temperature rise becomes smaller and the curve becomes flatter. The change near $z = 1$ also gets sharper, so a smaller h is needed to avoid oscillations.

4 Two-dimensional elliptic heat equation

The problem is

$$-\Delta T = f \quad \text{in } (0, 12) \times (0, 5),$$

with $T(x, 0) = T_{\text{ext}} = 25$. The left, right and top sides are insulated, so the derivative across each side is zero. The mesh has

$$h = \frac{12}{N} = \frac{5}{M}.$$

The values at the bottom are known. The other grid points, with $i = 0, \ldots, N$ and $j = 1, \ldots, M$, are numbered by

$$k = i + 1 + (j - 1)(N + 1).$$

Each numbered point gives one row in the matrix. The matrix size is therefore $(N+1)M \times (N+1)M$. At an inner grid point,

$$4T_{i,j} - T_{i-1,j} - T_{i+1,j} - T_{i,j-1} - T_{i,j+1} = h^2 f(x_i, y_j).$$

For $j = 1$, the known lower value adds T_{ext} to the right-hand side. The second-order boundary rows are

$$\begin{aligned}
-3T_{0,j} + 4T_{1,j} - T_{2,j} &= 0, \\
3T_{N,j} - 4T_{N-1,j} + T_{N-2,j} &= 0, \\
3T_{i,M} - 4T_{i,M-1} + T_{i,M-2} &= 0.
\end{aligned}$$

At the two upper corners we use the side rows. We now have one equation for each unknown.

4(a) Numerical solution for a constant source

For $f \equiv 2$ and $h = 0.2$, the computed value is

$$\boxed{T_h(6,2) = 41.000000000000}.$$

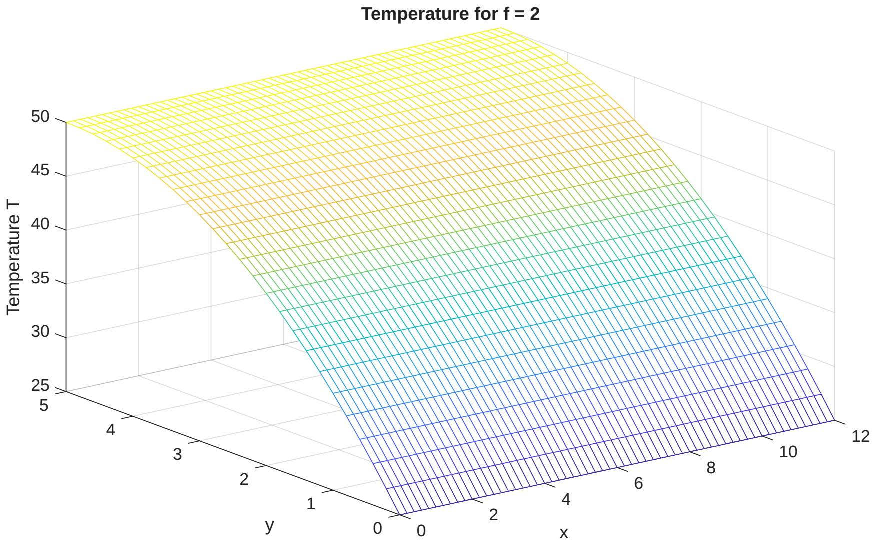

Figure 9: Numerical solution for $f \equiv 2$ and $h = 0.2$.

4(b) Exact solution for a constant source

Put $T = c_0 + c_1 y + c_2 y^2$ into the PDE and the boundary conditions. This gives

$$-2c_2 = 2, \qquad c_0 = 25, \qquad c_1 + 10c_2 = 0.$$

Hence

$$\boxed{T(x,y) = 25 + 10y - y^2}, \qquad T(6,2) = 41.$$

4(c) Why the numerical solution is exact

A Taylor expansion gives

$$\frac{g(x+h) - 2g(x) + g(x-h)}{h^2} = g''(x) + \frac{h^2}{12} g^{(4)}(x) + O(h^4).$$

For a quadratic, $g^{(3)} = g^{(4)} = 0$. The inner stencil and the one-sided boundary formulas are then exact. The difference at $(6,2)$ was only 3.70×10^{-13}, which comes from rounding in the computer.

4(d) Localized source

For

$$f(x,y) = 100 \exp\left(-\frac{1}{2}(x-4)^2 - 4(y-1)^2\right),$$

we build the matrix in the same way. Only the right-hand side changes.

Table 5: Localized-source values at $(6, 2)$.

h	N	M	$T_h(6, 2)$
0.20	60	25	47.219954048124
0.10	120	50	47.224023288355
0.05	240	100	47.224999298104

The successive-difference estimate is

$$\boxed{p = 2.059792},$$

which is close to order 2. The finest grid has 24100 unknowns. A sparse matrix saves both memory and time because most entries are zero.

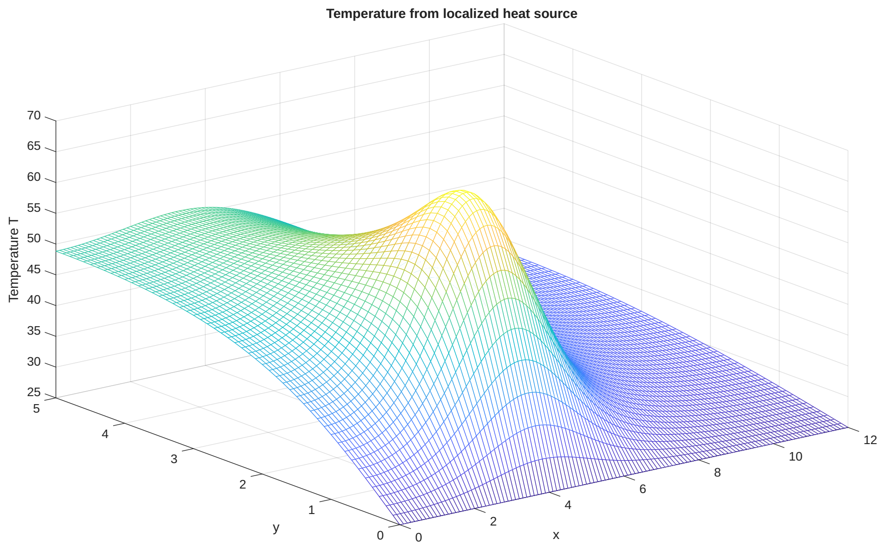

Figure 10: Mesh plot for the localized source with $h = 0.1$.

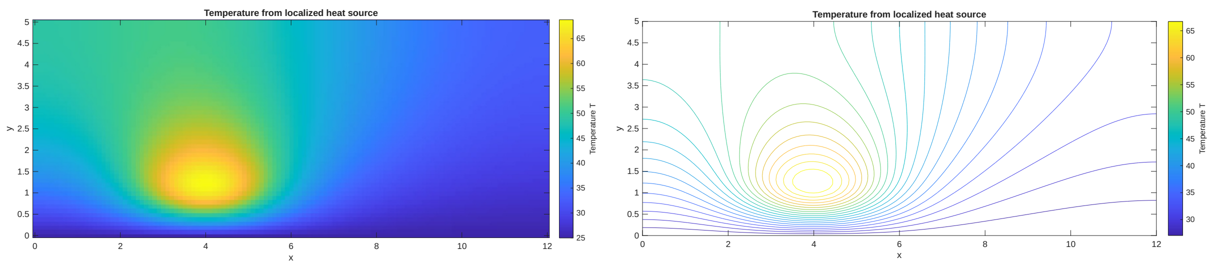

Figure 11: Image and contour plots for $h = 0.1$. The hottest area is close to the source at $(4, 1)$.